

Syllabus : Quadratic Equation

SECTION - I
Multiple Choice Questions (MCQ)

 This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- If the equation $x^2 + ax + b = 0$ has one root equal to unity and other root lies between the roots of the equation $x^2 - 7x + 12 = 0$ then range of values of a is
 (1) $(-5, -4)$ (2) $(-4, -3)$ (3) $(-3, -2)$ (4) $(4, 5)$
- Let $f(x) = ax^2 + bx + c$ where $a \neq 0$. If quadratic equation $f(x) = 0$ has positive distinct roots reciprocal to each other then
 (1) $af(1) > 0$ (2) $af(1) < 0$
 (3) $f(1) = 0$ (4) Nothing can be said about $af(1)$
- If $(x+1)(x+2)(x+3)(x+6) = 3x^2$, then the equation has
 (1) all imaginary roots (2) two imaginary and two rational roots
 (3) all rational roots (4) two imaginary and two irrational roots
- If the quadratic equation $x^2 - 4px + 4p(p-1) = 0$ possess roots of opposite sign then set of values of p is
 (1) $(0, 1)$ (2) $(-\infty, 0)$ (3) $(1, 4)$ (4) $(1, \infty)$
- If $x = \cos 15^\circ$, then which of the following is correct?
 (1) $4x^2 - 2 + \sqrt{3} = 0$ (2) $8x^2 - 6x + \sqrt{2} = 0$ (3) $8x^2 + 6x - \sqrt{2} = 0$ (4) $16x^4 - 16x^2 + 1 = 0$
- The quadratic equation $x^2 - 6x + 8 + \lambda(x^2 - 4x + 3) = 0$, $\lambda \in \mathbb{R}$ has
 (1) real and unequal roots for $\lambda \in \mathbb{R} - \{-1\}$ (2) real roots for $\lambda < 0$ only
 (3) real roots for $\lambda > 0$ only (4) real and unequal roots for $\lambda \in \mathbb{R}$
- If the ratio of roots of quadratic equation $\lambda x^2 + \mu x + \nu = 0$ is equal to the ratio of the roots of equation $x^2 + x + 1 = 0$ then λ, μ, ν are in
 (1) AP (2) GP (3) AGP (4) none of these
- If α, β, γ are roots of equation $x^3 - 2x^2 - 1 = 0$ and $T_n = \alpha^n + \beta^n + \gamma^n$ then the value of $\frac{T_{11} - T_8}{T_{10}}$ is equal to
 (1) 1 (2) 2 (3) -1 (4) 3
- If $\frac{1}{\sqrt{\alpha}}$ and $\frac{1}{\sqrt{\beta}}$ are the roots of the equation, $ax^2 + bx + 1 = 0$ ($a \neq 0, b \in \mathbb{R}$), then the equation, $x(x + b^3) + (a^3 - 3abx) = 0$ has roots
 (1) $\alpha^{3/2}$ and $\beta^{3/2}$ (2) $\alpha\beta^{1/2}$ and $\alpha^{1/2}\beta$ (3) $\sqrt{\alpha\beta}$ and $\alpha\beta$ (4) $\alpha^{-3/2}$ and $\beta^{-3/2}$
- The set of values of α for which the quadratic equation $(\alpha+2)x^2 - 2\alpha x - \alpha = 0$ has two roots on the number line symmetrically placed about the point 1 is

- (1) $\{-1, 0\}$ (2) $\{0, 2\}$ (3) ϕ (4) $\{0, 1\}$
11. The values of k for which each root of the equation, $x^2 - 6kx + 2 - 2k + 9k^2 = 0$ is greater than 3, always satisfy the inequality:
 (1) $7 - 9k > 0$ (2) $11 - 9k < 0$ (3) $29 - 11k > 0$ (4) $29 - 11k < 0$
12. For real x , the value of expression $\frac{x^2 - 2x + 2}{2x - 2}$ cannot lie in the interval
 (1) $(-\infty, 1)$ (2) $(-1, 1)$ (3) $(1, 2)$ (4) $(2, \infty)$
13. If $c^2 = 4d$ and the two equations $x^2 - ax + b = 0$ and $x^2 - cx + d = 0$ have a common root, then the value of $2(b + d)$ is equal to
 (1) $\frac{a}{c}$ (2) ac (3) $2ac$ (4) $a + c$
14. Let $P(x) = x^2 - (2 - p)x + (p - 2)$. If $P(x)$ assumes both positive and negative values $\forall x \in \mathbb{R}$, then the range of values of p is
 (1) $(-\infty, 2) \cup (6, \infty)$ (2) $(2, 6)$ (3) $(-\infty, 2)$ (4) $(6, \infty)$
15. Find range of values of c for which range of expression $\frac{2x + c}{x^2 + x - c}$ is $(-\infty, \infty)$
 (1) $[0, 6]$ (2) $(0, 6)$ (3) $[0, 8]$ (4) $(0, 8)$
16. If $0 < a < b < c$ and the roots α, β of the equation $ax^2 + bx + c = 0$ are imaginary then –
 (1) $|\alpha| = |\beta| > 1$ (2) $|\alpha| = |\beta| < 1$
 (3) $|\alpha| \neq |\beta|$ (4) None of these
17. If the equation $x^2 - 3px + 2q = 0$ and $x^2 - 3ax + 2b = 0$ have a common roots and the other roots of the second equation is reciprocal of the other roots of the first then $(2q - 2b)^2$ equals-
 (1) $36 pa (q - b)^2$ (2) $18 pa (q - b)^2$
 (3) $36 bq (p - a)^2$ (4) $18 bq (p - a)^2$
18. If both the roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then k lies in the interval
 (1) $(5, 6]$ (2) $(6, \infty)$ (3) $(-\infty, 4)$ (4) $[4, 5]$
19. If x is real and $y = \frac{x^2 - x + 3}{x + 2}$, then
 (1) $y \geq 10$ (2) $y \geq 11$ (3) $y \leq -11$ or $y \geq 1$ (4) $-11 < y < 1$
20. If the equation $ax^2 + 2bx - 3c = 0$ have no real roots and $4(a + b) > 3c$ then
 (1) $c < 0$ (2) $c > 0$ (3) $c \geq 0$ (4) $c = 0$

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. If $\alpha, \beta \in \mathbb{R}$ and the quadratic equations $x^2 + 2x + 7 = 0$ and $4x^2 + \alpha x + \beta = 0$ have a common root, then find the value of $\alpha + \beta$
22. If equation $x^2 + ax + a + 1 = 0$ has integral roots then find the sum of possible integral values of 'a'
23. If $a, b \in \{1, 2, 3, 4\}$ and $ax^2 + bx + 1 = 0$ has real roots then find the number of possible pairs (a, b)

24. If α, β are roots of $375x^2 - 25x - 2 = 0$ and $S_n = \alpha^n + \beta^n$, then find $\lim_{n \rightarrow \infty} \sum_{r=1}^n S_r$
25. Number of real solutions of $|x| + 2\sqrt{5 - 4x - x^2} = 16$ is /are

Answer Key_DPP-02

1.	(1)	2.	(2)	3.	(4)	4.	(1)	5.	(4)	6.	(1)	7.	(2)
8.	(2)	9.	(1)	10.	(3)	11.	(2)	12.	(2)	13.	(2)	14.	(1)
15.	(2)	16.	(1)	17.	(3)	18.	(3)	19.	(3)	20.	(1)	21.	(36)
22.	(4)	23.	(7)	24.	(0.08)	25.	(0)						